Quantitative Easing and the Balance-Sheet Channel: An Econometric Analysis of Term Premium Compression in the United States

Anthony Rivera

ECON 5283 – Data Visualization with Python

December 2025

Abstract

This paper examines how the Federal Reserve's large-scale asset purchases, commonly referred to as quantitative easing (QE), affect long-maturity U.S. Treasury yields through the balance-sheet (portfolio-balance) channel. Using the Adrian–Crump–Moench (ACM) 10-year term premium as the primary dependent variable and System Open Market Account (SOMA) holdings as the central policy instrument, the analysis combines descriptive statistics, regression methods with heteroskedasticity- and autocorrelation-consistent inference, and vector autoregression techniques. The results indicate that QE influences the term premium dynamically, with balance-sheet expansions compressing term premia over several months and accounting for a meaningful share of its forecast variance. These findings support the preferred-habitat view of bond markets and highlight QE's role as a powerful but transitory policy tool during periods when conventional interest-rate policy is constrained.

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1 Introduction

Quantitative easing has become one of the Federal Reserve's most prominent unconventional monetary policy tools, particularly during periods when short-term policy rates are constrained by the zero lower bound. By purchasing long-maturity Treasury securities and agency mortgage-backed securities, the Federal Reserve directly alters the composition of assets held by the private sector. According to the balance-sheet or portfolio-balance channel, these purchases reduce the supply of long-duration securities available to investors, thereby compressing the term premium embedded in long-term interest rates.

This paper investigates whether and how QE operates through this balance-sheet channel in the United States. Using updated data through 2025, the analysis evaluates both the static and dynamic relationship between the Federal Reserve's asset holdings and the 10-year Treasury term premium. The study contributes to the existing literature by incorporating post-pandemic QE episodes and applying a unified empirical framework that combines regression analysis with vector autoregression, impulse response functions, and forecast error variance decomposition.

The remainder of the paper proceeds as follows. Section 2 outlines the research question and motivation. Section 3 describes the data and variable construction. Section 4 presents descriptive evidence on the evolution of the Federal Reserve's balance sheet. Section 5 details the empirical strategy. Section 6 presents the econometric results. Section 7 discusses the broader policy implications, and Section 8 concludes.

2 Research Question and Motivation

2.1 Research Question

How have the Federal Reserve's large-scale asset purchases affected long-maturity U.S. Treasury yields through the balance-sheet channel, specifically by reducing the market's available duration and compressing the term premium?

2.2 Motivation

QE has played a central role in U.S. monetary policy since the Global Financial Crisis and again during the COVID-19 pandemic. While conventional monetary policy operates primarily through expectations of future short-term interest rates, QE is intended to influence long-term rates directly by altering the supply of assets available to private investors. Understanding whether this mechanism operates through the balance-sheet channel is crucial for evaluating the effectiveness of QE and for informing future policy design, particularly with respect to maturity targeting and balance-sheet normalization.

This question is academically relevant because it speaks to competing theories of the term structure, including the preferred-habitat hypothesis, and policy relevant because it informs debates about the long-run role of central bank balance sheets.

3 Data

3.1 Data Sources

The analysis uses monthly data drawn from the following sources:

- Federal Reserve Bank of New York: System Open Market Account (SOMA) holdings
- Adrian–Crump–Moench (ACM): 10-year Treasury term premium estimates
- U.S. Treasury: Total Treasury debt outstanding
- Federal Reserve Economic Data (FRED): CPI inflation, federal funds rate, and VIX

The sample is restricted to the modern QE era beginning in the early 2000s and focuses primarily on the post-2020 period for dynamic analysis.

3.2 Variable Construction

The primary dependent variable is the ACM 10-year term premium (ACMTP10). QE intensity is proxied using both the level of total SOMA holdings and SOMA holdings as a share of outstanding Treasury debt. Control variables include total Treasury debt outstanding, the effective federal funds rate, year-over-year CPI inflation, and the VIX as a measure of market volatility.

4 Descriptive Evidence

4.1 Summary Statistics

Table 1: Summary Statistics: SOMA Asset Holdings (Levels)

Asset Class	Count	Mean	Std. Dev.	Min
Total	1.161×10^3	3.605×10^{12}	2.440×10^{12}	4.733×10^{11}
MBS	1.161×10^3	1.257×10^{12}	8.973×10^{11}	0.000
TIPS	1.161×10^3	1.350×10^{11}	1.252×10^{11}	1.281×10^{10}
FRN	5.940×10^2	1.330×10^{10}	8.884×10^9	0.000
TIPS Inflation Compensation	1.161×10^3	2.953×10^{10}	3.661×10^{10}	9.287×10^8
Notes & Bonds	1.161×10^3	2.044×10^{12}	1.426×10^{12}	3.989×10^{11}
Bills	1.161×10^3	1.306×10^{11}	1.327×10^{11}	0.000
Agencies	9.130×10^2	3.830×10^{10}	4.932×10^{10}	0.000
CMBS	1.161×10^3	2.148×10^9	3.765×10^9	0.000

The descriptive statistics provide an initial quantitative portrait of the Federal Reserve's SOMA portfolio. Mean total holdings exceed $3.6 trillion, with substantial dispersion reflecting the episodic expansion of the balance sheet during QE programs. Notes and bonds constitute the largest component, highlighting the emphasis on long-duration securities.

The distributional properties reveal wide variation and non-normality across asset categories, consistent with discrete and policy-driven purchase episodes rather than smooth accumulation.

4.2 Growth Rates

Table 2: Weekly Growth Statistics of SOMA Components (%)

Series	Count	Mean (%)	Std. Dev.	Min (%)
Total Growth	1160.000	0.203	1.389	−7.354
MBS Growth	873.000	inf	NaN	−2.489
TIPS Growth	1160.000	0.284	1.390	−6.641
FRN Growth	583.000	inf	NaN	−40.962
TIPS Inflation Comp. Growth	1160.000	0.470	3.583	−36.917
Notes & Bonds Growth	1160.000	0.193	0.905	−2.247
Bills Growth	821.000	inf	NaN	−100.000
Agencies Growth	911.000	inf	NaN	−100.000
CMBS Growth	288.000	inf	NaN	−1.211

Weekly growth rates of SOMA components display substantial volatility, with large spikes corresponding to the initiation and scaling of QE programs. Several asset categories exhibit extreme or undefined growth rates due to zero holdings in earlier periods, underscoring the episodic nature of QE implementation.

4.3 Portfolio Composition Ratios

Table 3: Average SOMA Portfolio Composition

Asset Class	Average Share (%)
Notes & Bonds	57.52
MBS	28.44
TIPS	3.51

On average, Notes and Bonds account for approximately 58% of total SOMA holdings, followed by MBS at roughly 28%. TIPS represent a comparatively small share. This composition confirms that QE primarily targets duration-sensitive assets, consistent with the balance-sheet channel.

4.4 Correlation Patterns

High positive correlations among major SOMA asset categories indicate that balance-sheet expansions were broad-based rather than narrowly targeted. These correlations caution against simple bivariate inference and motivate multivariate econometric analysis.

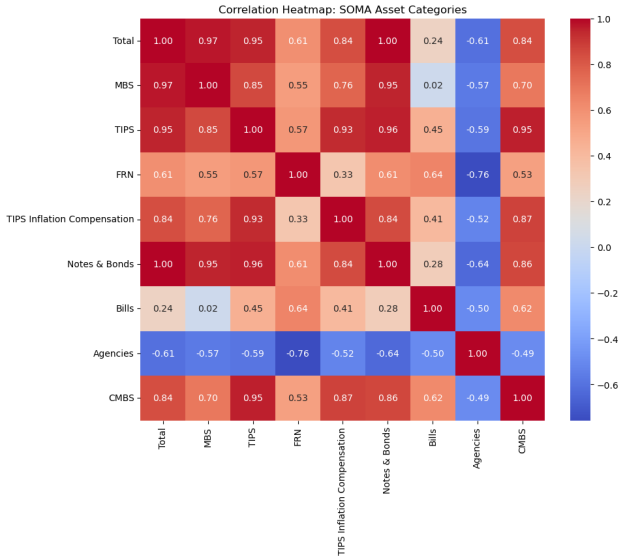


Figure 1: Correlation Heatmap: SOMA Asset Categories

4.5 Time-Series Evolution

The time-series evolution of SOMA holdings shows pronounced increases during QE1–QE3 and the 2020 pandemic response, followed by plateau and runoff phases. These patterns highlight both the scale and reversibility of balance-sheet interventions.

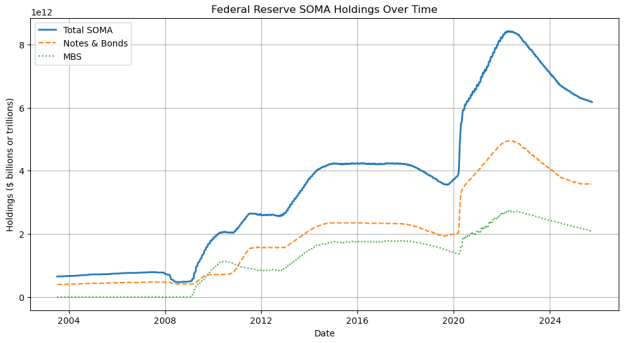


Figure 2: Federal Reserve SOMA Holdings Over TimeI

4.6 Interim Summary

Taken together, the descriptive evidence demonstrates that QE materially altered both the size and composition of the Federal Reserve's balance sheet, with a persistent tilt toward long-duration assets. This behavior aligns with the theoretical prediction that QE compresses term premia by absorbing duration risk from private investors.

5 Empirical Strategy

5.1 Baseline Regression

The baseline specification relates the 10-year term premium to contemporaneous and lagged measures of QE intensity:

$$ACMTP10_t = \alpha + \beta_1 SOMA_t + \beta_2 SOMA_{t-1} + \beta_3 SOMA_Share_t + \beta_4 SOMA_Share_{t-1} + \gamma' X_t + \varepsilon_t,$$

where X_t includes Treasury supply, the federal funds rate, inflation, and market volatility.

Inference is conducted using Newey–West HAC standard errors to account for serial correlation and heteroskedasticity.

5.2 Dynamic Framework

Given evidence of residual autocorrelation, a vector autoregression (VAR) framework is employed to capture the dynamic interactions among the term premium, QE measures, and macro-financial controls. Impulse response functions and forecast error variance decomposition are used to evaluate the timing and persistence of QE effects.

6 Empirical Results

6.1 OLS Results

The HAC-corrected OLS estimates indicate that the level of SOMA holdings is not statistically significant once controls are included, while the lagged share of SOMA holdings in the Treasury market exhibits a negative and statistically significant relationship with the term premium. This suggests that QE effects materialize with a delay, consistent with portfolio rebalancing behavior.

Table 4: OLS Estimates with HAC Standard Errors

Variable	Coefficient	Std. Error	p-value
SOMA_Total	-5.52×10^{-13}	1.63×10^{-12}	0.736
SOMA_Total$_{t-1}$	7.02×10^{-13}	1.30×10^{-12}	0.592
SOMA_Share	0.3518	0.162	0.034
SOMA_Share$_{t-1}$	-0.4393	0.154	0.006
Total Debt Outstanding	5.48×10^{-14}	3.17×10^{-14}	0.090
Fed Funds Rate	-0.0920	0.056	0.106
CPI YoY	0.0585	0.149	0.697
VIX	-0.0071	0.009	0.407

6.2 Diagnostics

Durbin–Watson and Ljung–Box tests strongly reject the null of no serial correlation, confirming the need for a dynamic specification.

6.3 VAR and Impulse Responses

The VAR results show strong persistence in the term premium and balance-sheet variables. Impulse response functions reveal that a positive QE shock leads to a statistically significant decline in the term premium over a 2–4 month horizon, followed by gradual mean reversion.

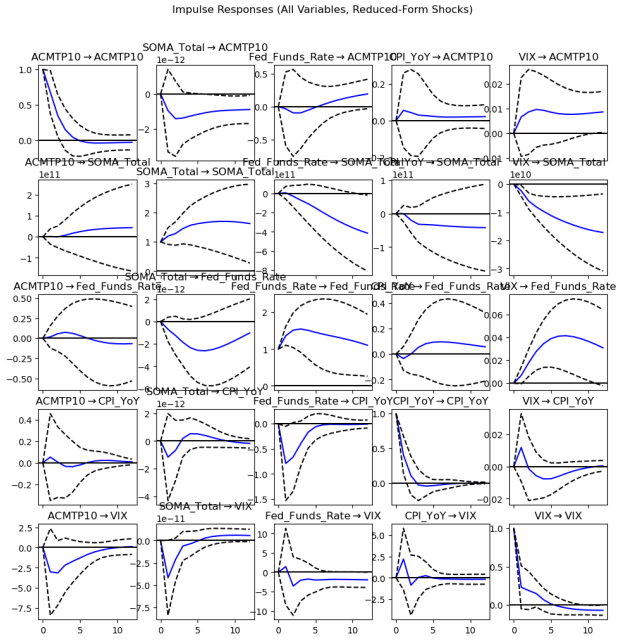


Figure 3: Impulse Responses (All Variables)

6.4 Forecast Error Variance Decomposition

Forecast error variance decomposition indicates that QE shocks explain approximately 10% of the term premium's variance at a 12-month horizon, comparable in magnitude to the contribution of market volatility.

7 Discussion and Policy Implications

Table 5: Summary of Variance Decomposition Results

Equation (Dependent Variable)	Main Source of Variation	QE (SOMA_Total) Share	Interpretation
ACMTP10 (Term Premium)	Own history + SOMA shocks	∼11%	QE explains a meaningful share of long-term yield dynamics through the balance-sheet channel.
SOMA_Total (QE)	Self + VIX	∼10–30% from VIX	QE responds endogenously to financial stress, confirming its countercyclical use.
Fed_Funds_Rate	SOMA + Self	∼39% from SOMA	Short-rate expectations co-move with QE actions, reflecting policy coordination.
CPI_YoY	Self	∼3%	Inflation expectations remain largely autonomous from balance-sheet policy.
VIX	Self + SOMA	∼11% from SOMA	QE dampens financial market volatility via liquidity provision and confidence effects.

The results support the balance-sheet channel of QE, demonstrating that asset purchases compress the term premium dynamically rather than instantaneously. The findings imply that QE is an effective stabilization tool during periods of market stress, but its effects are transitory and decay as markets adjust.

From a policy perspective, these results suggest that maintaining large balance sheets indefinitely may yield diminishing returns and increase the risk of market distortions.

8 Conclusion

This paper provides empirical evidence that the Federal Reserve's balance-sheet expansions reduce the U.S. Treasury term premium through the portfolio-balance channel. While QE effects are economically meaningful, they are temporary and operate primarily through dynamic adjustment rather than immediate price changes. Future research should explore alternative identification strategies and longer samples to further assess the robustness of balance-sheet transmission mechanisms.

References

A Appendix

Additional robustness checks, figures, and tables may be placed here.

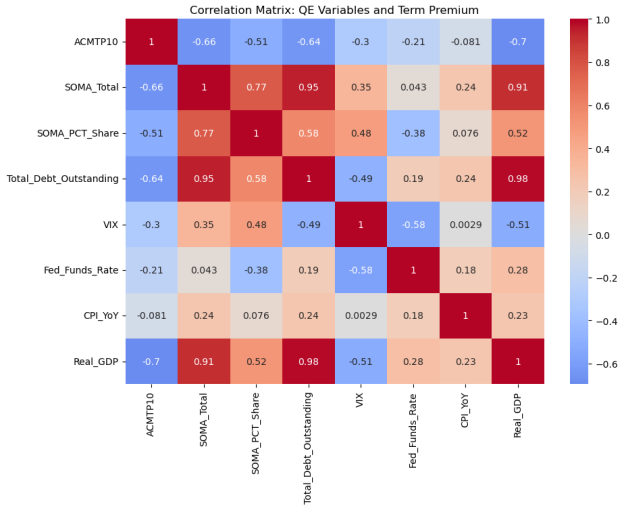


Figure 4: Correlation Matrix: QE Variables Term Premium

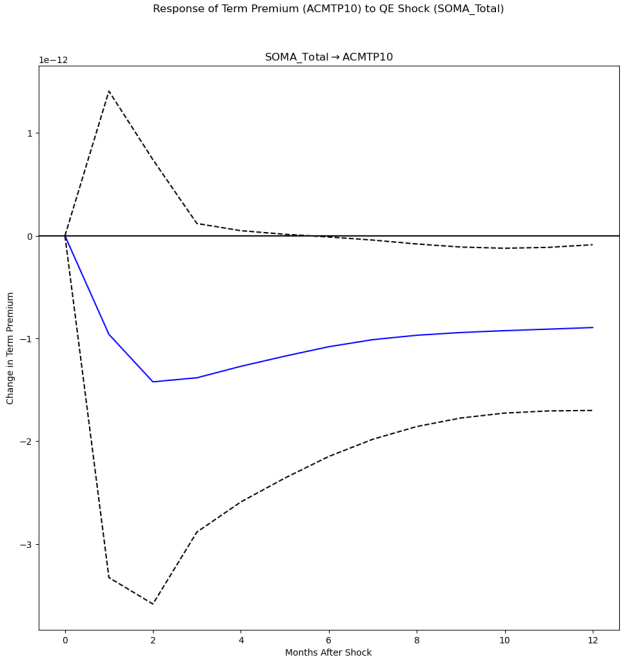


Figure 5: Response of Term Premium (ACMTP10) to QE Shock (Soma$_Total$)

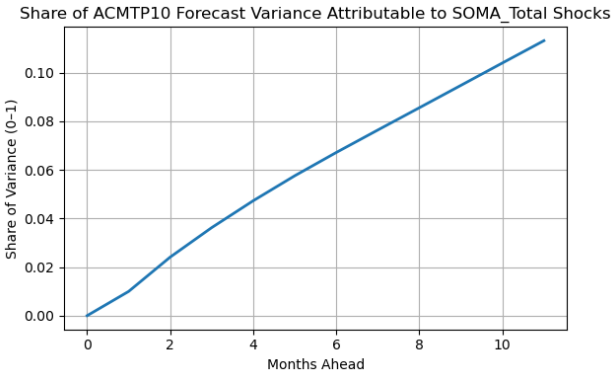


Figure 6: Share of ACMTP10 Variance Attributable to SOMA$_{T}otalShocks$